Ruprecht-Karls-Universität Heidelberg
Fakultät für Ingenieurswissenschaften

Studying unique CpG methylation patterns in BRCA pathogenesis

Data analysis project report

Autorinnen Antonia Eisenhofer, Irena Dikic, Kathrin Kreischer, Marie Sanders
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Abstract

Breast cancer (BRCA) remains a leading cause of mortality among women worldwide, with early detection being critical for improving survival rates. Epigenetic mechanisms, particularly DNA methylation changes at CpG sites, play a crucial role in BRCA pathogenesis. This project investigates unique CpG methylation patterns associated with BRCA using data from the Cancer Genome Atlas generated by Illumina Infinium Human Methylation 450 Bead Chip array. Differential methylation analysis (DMA) revealed significant differences in CpG sites between BRCA and normal tissues, explored using Principal Component Analysis, Uniform Manifold Approximation and Projection, correlation analysis and hierarchical clustering. Additionally, the most significant genes were identified via DMA and linked to critical pathways implicated in tumorigenesis and metastasis using hallmark gene set enrichment analysis. These pathways include epithelial-mesenchymal transition, myogenesis, apical junction, the KRAS cascade, inflammatory response, IL-2/STAT5 and IL-6/STAT3 signalling. Most importantly, the notable dysregulation of estrogen response pathways highlight their crucial relevance in BRCA pathogenesis. These findings emphasize the pivotal role of epigenetic mechanisms, particularly methylation patterns, in refining diagnostic strategies and potentially advancing personalized treatment approaches for BRCA.

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Abbreviations

BRCA	Breast cancer
DMA	Differential Methylation Analysis
DMR	Differential Methylated Region
DMP	Differentially methylated position
DMS	Differentially methylated sites
EMT	Epithelial-Mesenchymal Transition
ER	Estrogen receptor-α
MSigDB	Molecular Signature Database
PC	Principal Component
PCA	Principal Component Analysis
TSG	Tumor Supressor Gene
TCGA	The Cancer Genome Atlas
UMAP	Uniform Manifold Approximation and Projection

1 Introduction

In 2022, 2.3 million women were diagnosed with breast cancer (BRCA), leading to 670,000 deaths from the disease. Breast cancer is present worldwide, impacting women of all ages after puberty (World Health Organisation, 2024). Early detection of BRCA, which significantly enhances treatment efficacy and plays a pivotal role in increasing survival rates, is therefore crucial in helping all patients (Zhang *et al.*, 2020).

Many genetic aberrations associated with breast cancer are well established, such as BRCA1 and BRCA2, and HER-2 (Honrado *et al.*, 2006). Only in recent years has there been a notable increase in research focusing on the significance of epigenetic mechanisms, particularly DNA methylation and its influence on BRCA development.

DNA methylation primarily occurs in DNA sequences where cytosine nucleotides are followed by guanine nucleotides, known as CpG sites. In these sequences, cytosine is methylated at the 5' carbon position by DNA methyltransferase enzymes. Regions of the eukaryotic genome with a high density of CpG sites are called CpG islands. Approximately 60% of gene promoters are located within CpG islands, including critical housekeeping genes essential for cellular processes and tumor supressor genes (TSGs) (Alberts, 2015; Besselink *et al.*, 2023).

The two main patterns of DNA methylation associated with cancer are genome-wide hypomethylation and hypermethylation of CpG islands in promoter regions (Cheng *et al.*, 2018). Hypermethylation of CpGs in promoter regions can directly inhibit transcription factor binding, activate specific trascriptional repressors or inactivate chromatin structure formation, all leading to gene silencing. These mechanisms could result in the silencing of TSGs, thereby promoting tumor development (Singal and Ginder, 1999).

DNA hypomethylation mainly involves the demethylation of typically methylated DNA repetitive sequences. This process has been linked to chromosomal instability, a major driver of tumor evolution characterized by errors in chromosome segregation during mitosis (Endo *et al.*, 2022).

Bisulfite conversion is the most common technique for determining methylation status. In this method, DNA is treated with sodium bisulfite, which converts unmethylated cytosine to uracil, and through PCR to thymine, while methylated cytosine remains unaffected. Following bisulfite treatment, technologies like the HumanMethylation450 BeadChip array enable efficient

and quantitative assessment of selected methylation sites across the genome. These measurements are subsequently transformed into β-values, representing the proportion of methylation at a specific CpG site within the sample (Miyata *et al.*, 2017; Illumina, 2012).
The diagnostic, prognostic and therapeutic importance of methylation patterns is currently an area of active research. In this project, we aim to find specific methylation patterns in breast cancer pathogenesis by using Differental methylation analysis (DMA) to find significant methylational differences between BRCA and normal tissue. Those CpGs were used to identify diferentially methylated regions (DMRs). In order to connect the results to hallmark gene sets, gene set enrichment analysis was performed.

2 Material and Methods

2.1 Referenced Data

We were provided with methylation data from The Cancer Genome Atlas (TCGA). Sequencing took place through Illumina Infinium Human Methylation 450 Bead Chips.
Sequencing data is then extracted to β-values, which show the methylated proportion of a specific CpG site in the sample. The higher the β-value, the more cells from the sample are methylated in this site.
The methylation data consisted of 30 primary solid tumor samples and 30 normal solid tissue samples for 426,461 CpG sites. An additional sample sheet provides clinical information on the samples. Pre-processing of data was performed before the start of this project to remove certain CpGs.

2.2 Data Exploration

To explore relationships between samples or methylation levels, a correlation analysis was conducted. Specifically, Pearson correlation coefficients were computed to assess the linear relationship between methylation levels of probe pairs across samples. The correlation was visualized through heatmaps created using the `pheatmap` package in R. Through hierarchical agglomerative clustering with complete linkage and euclidean distances, groups of patients or CpGs with similar methylation patterns were identified. To visually explore the distribution of β-values and compare this distribution among patients with different tissue types, box plots were generated using the `ggplot2` package in R.
Principal component analysis (PCA) was performed using the function `prcomp()` to reduce the dimension of the original dataset into a set of orthogonal components known as principal components (PCs).

2.3 Differential Methylation Analysis

To identify differentially methylated positions (DMPs) between sample groups, two distinct methods were employed: the `limma` package from Bioconductor and the `dmpFinder` function from the `minfi` package. The `limma` (Linear Models for Microarray Data) package fits linear models to the methylation data. The `dmpFinder` function employs a moderated t-statistic to test for differential methylation.
The input consisted of all CpG sites from the 60 patients included in the dataset. All CpGs with an adjusted p-value/q-value below 0.05 were considered significant. For downstream analysis, the CpGs significant in both methods were used.
Using the `umap` package in R, we embedded the 426,461 CpGs into a two-dimensional space using the Euclidean distance metric and considering 15 neighboring sample points for manifold approximation. The 160,587 significant CpGs from the DMA results were marked. The UMAP plot was generated using the `ggplot2` package in R.

2.4 Differentially Methylated Region Analysis

Differentially methylated regions (DMRs) were identified using the 5000 most significant CpGs from the DMA results. For this, the `DMRcate` package was used which clusters significant CpG sites and calculates a combined test statistic to identify regions that exhibit a significant aggregated difference. For visualization, the `Gviz` package was employed. This package provides a comprehensive framework for plotting genomic data along genomic coordinates by utilizing different tracks that are integrated to form a complete plot. We utilized the `GenomeAxisTrack` and the `IdeogramTrack` to reference the genomic location on a chromosome. Additionally, `DataTrack` was used to visualize the methylation levels of the CpG sites comprising the DMR, and the `AnnotationTrack` was used to define the DMR region.

2.5 Methods for literature comparison

2.5.1 Comparison of methylation status with diagnostic markers of Zhang *et al.*

In the 2020 study by Zhang *et al.*, seven diagnostic marker CpG sites for BRCA were identified through analysis of differential methylation patterns in BRCA tumors, adjacent normal tissues, and healthy blood. The study employed filtering of differentially methylated sites (DMSs) between BRCA and healthy blood, cross-referenced these findings with nine other common cancers, and selected representative DMSs using the WrapperSubsetEval evaluator. To assess whether these diagnostic markers exhibit differential methylation in the data operated on in this project, the significance of β-value values between normal and tumor tissue was examined at seven CpG sites (cg20383521, cg09804858, cg06741896, cg01668352, cg10708955, cg06998282, cg04658021). The Boxplots of these CpGs were generated using the `ggpubr` package in R and significance computed by the DMA was annotated using asterisks. The correlation between the seven CpGs was computed using `pheatmap` and the color added using `ColorBrewer`. Hierarchical clustering was computed using complete linkage and euclidean distances.

2.5.2 Comparison of DMR results with genes identified by Panagopoulou *et al.*

To validate the DMR results with existing literature, a comparison was made with the top 250 ranking genes identified by Panagopoulou *et al.* in 2021. They utilized the `RnBeads` R package to rank genes that were differentially methylated between BRCA and normal tissue. The DMR results, including all 11,782 identified genes, were compared with the top 250 genes from their study. Additionally, the intersection of the top 500 genes from the DMR result with their top 250 genes was examined. Overlapping genes between both datasets were visualized using Venn Diagrams generated with the R package `VennDiagram`.

2.6 Hallmark gene enrichment analysis

To conduct hallmark gene enrichment analysis for the significant CpGs identified from the DMA, the `missMethyl` package was utilized.
This package was chosen to mitigate biases associated with interpreting annotated genes

manually. Specifically, it addresses the issue where genes with more CpG loci are more likely to be identified as differentially methylated, and the problem of different genes sharing CpG loci, which can lead to pathways being incorrectly categorized as differentially methylated.
The hallmark gene set required for this analysis was downloaded from the Molecular Signature Database (MSigDB) (Liberzon *et al.*, 2015).
The results were visualized in a bar plot using the `ggplot2` package.

2.7 Packages used

Table 2.1 List of the main packages used, their versions and sources.

Package/Algorithm	Version	Source
ggplot2	3.5.1	(Wickham, 2016)
pheatmap	1.0.12	(Kolde, 2019)
umap	0.2.10.0	(McInnes *et al.*, 2018)
limma	3.60.3	(Alberts, 2015)
minfi	1.50.0	(Aryee *et al.*, 2014)
Gviz	1.46.1	(Hahne and Ivanek, 2016)
DMRcate	2.16.1	(Peters *et al.*, 2015)
missMethyl	1.36.0	(Maksimovic *et al.*, 2012)
VennDiagramm	1.7.3	(Chen and Boutros, 2011)
ColorBrewer	1.1-3	(Brewer *et al.*, 2003)

3 Results

3.1 Patients cluster into two groups

Before taking a closer look at the methylation data itself, it was examined whether the patients showed a distinct clustering pattern. The most decisive factor that divides the patients is their tissue type (primary solid tumor vs. normal solid tissue). According to this classification the patients were colored in blue for solid tissue and red for tumor tissue.
In the colored PCA it can be seen that there are two distinct clusters forming for the Primary solid Tumor and the Solid Tissue Normal. It is notable that two tumor samples cluster with the normal samples.
Furthermore it can be observed that the clustering of the patients for the solid tissue is more tight than for the tumor tissue (Fig. 3.1).

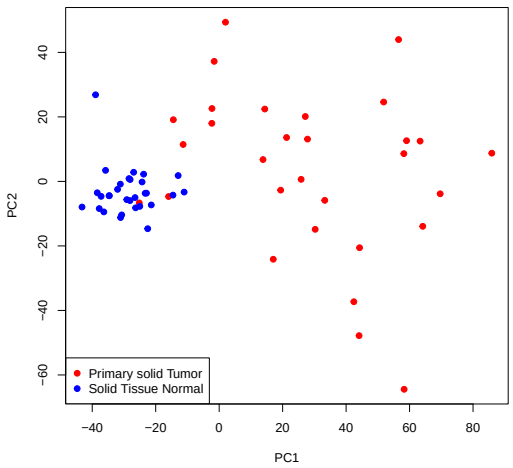


Figure 3.1: Plot of the first two princpal components from the PCA of the patients. Patients are colored according to their tissue type, red for Primary solid Tumor and blue for Solid Tissue Normal.

3.2 Identifying differentially methylated probes

The DMA resulted in a list of differently methylated CpGs. Subsequently, significant CpGs were selected based on the adjusted p-value or q-values with a significance level set to 0.05. The `minfi` package identified 202,725 significantly differentially methylated CpGs, while the `limma` package identified 160,587 significantly differentially methylated CpGs. For further analysis, the intersection of the results of the two methods was used, comprising the 160,587 CpGs from the `limma` method.
The six most significant CpGs are visualized as boxplots. Each CpG shows a distinct difference in methylation level for the two tissue types, with CpGs in tumor samples being hypermethylated in three CpG sites (cg11743000, cg16754788, cg00692173) and hypomethylated in three CpG sites (cg04786142, cg03130910, cg23811057)(Fig. 3.2).

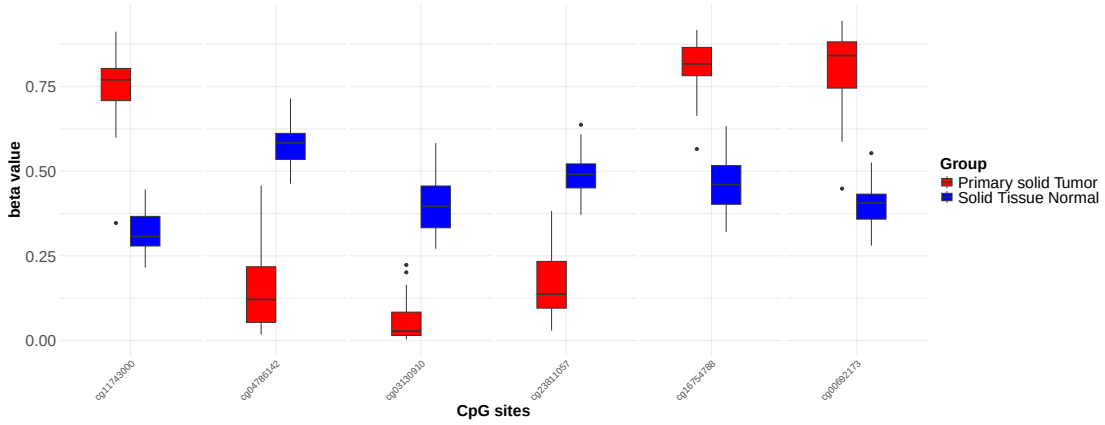


Figure 3.2: Boxplot of the six most significantly differentially methylated CpGs. The y-axis represents the β-values, while the x-axis lists CpGs. Tumor samples are colored in red and normal samples are colored in blue.

3.2.1 Significantly differentially methylated CpGs can be used as markers for the health status

The patients and various subgroups of the significantly differentially methylated CpGs were examined. For this purpose, a correlation analysis among the patients was conducted considering the top 500 CpGs, along with a PCA of the patients for the top 1000 CpGs. When examining the correlation heatmap and annotation based on Primary Solid Tumor and Solid

Tissue Normal types, it can be observed that tumor patients exhibit a high correlation with other tumor patients, while normal patients show strong correlation with other normal patients. Tumor patients and normal patients show a negative correlation. This pattern is also evident in the dendrogram, where patient groups can be divided into two large clusters (Fig. 3.3A). Upon examination of the PCA, two clearly distinct clusters can be identified. These can also be differentiated based on the coloring of Primary Solid Tumor and Solid Tissue Normal (Fig. 3.3B).

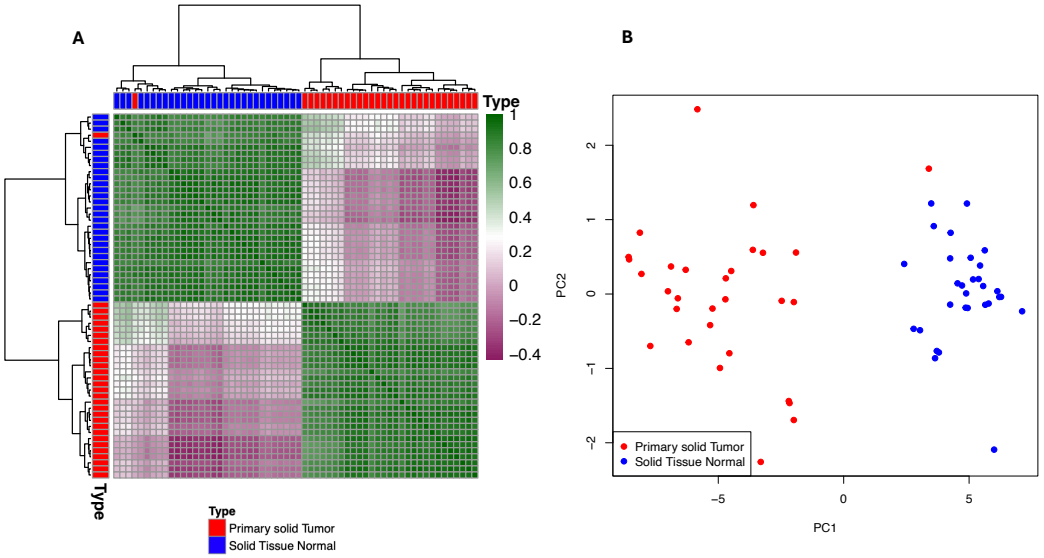


Figure 3.3: Visualization of patients based on the 1000 most significant CpGs. (A) Heatmap for correlation betweeen the 60 patients. Positive correlation is shown in green and negative correlation is shown in purple. Tumor patients are annotated in red and normal patients are annotated in blue. (B) PCA of patients. Tumor patients are colored in red and normal patients are colored in blue.

In both the PCA and the clustering in the correlation heatmap, it is noticeable that a patient with a Primary Solid Tumor profile is located within the cluster of patients with Solid Tissue Normal. Upon closer examination of the patient and in comparison with the 60 other patients, it was found that this patient does not stand out from the others clinically. The patient is female like 58 other patients, and based on the given characteristics in the sample sheet, it cannot be explained why this patient stands out (Table 3.1).

Table 3.1 Patient TCGA-AO-A1KO-01A-31D-A13K-05: Characteristics extracted from the given sample sheet.

TCGA.ID	TCGA-AO-A1KO-01A-31D-A13K-05
Type	Primary solid Tumor
Gender	FEMALE
Ethnicity	NOT HISPANIC OR LATINO
Race	WHITE

After the DMA, the CpGs were examined more closely. For this purpose, a correlation analysis and a PCA for the 1000 most significant CpGs were conducted. A clear pattern emerges when examining the heatmap of the correlation among the 1000 most significantly differentially methylated CpGs. Very high positive correlation is observed among hypermethylated and hypomethylated CpGs. Hypermethylated CpGs show a very high negative correlation with hypomethylated CpGs. Additionally, the dendrogram analysis reveals a clear separation, clustering CpGs into two distinct groups based on their hypermethylation or hypomethylation status (Fig. 3.4A). Similarly, the PCA analysis of these top 1000 CpGs demonstrates the formation of two well-defined and separate clusters, corresponding to hypermethylated and hypomethylated CpGs (Fig. 3.4B).

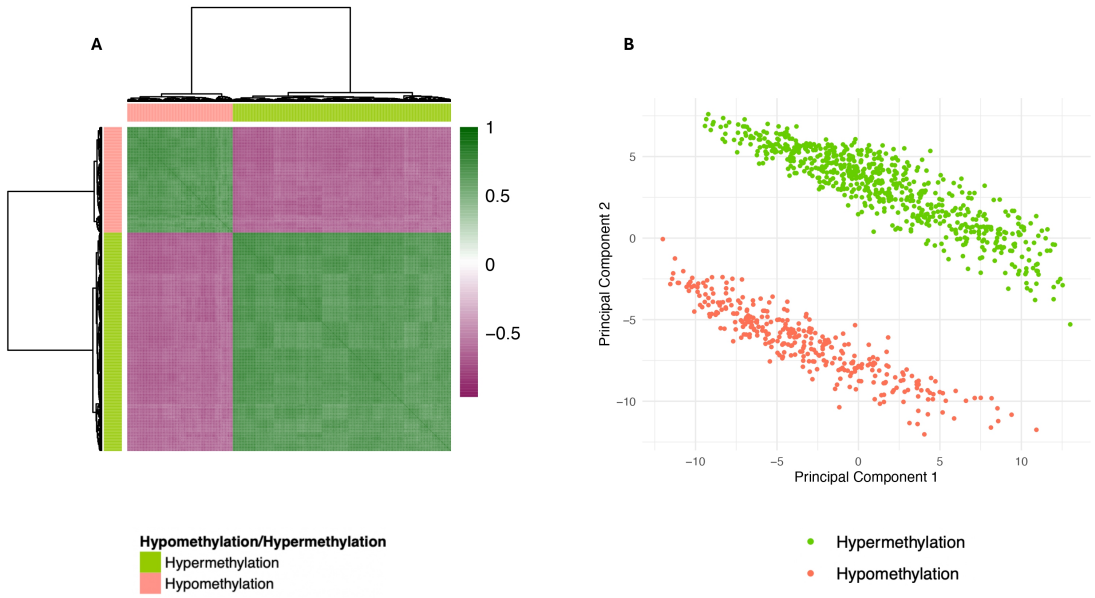


Figure 3.4: Visualization of the 1000 most significant CpGs. (A) Correlation heatmap for the CpGs. Positive Correlation is shown in green and negative correlation is shown in purple. Hypermethylation is annotated in green and Hypomethylation is annotated in salmon. CpGs are hierachical clustered. (B) PCA plot for the CpGs. Hypermethylation is annotated in green and hypomethylation is annotated in salmon.

3.2.2 UMAP analysis of CpG distribution

UMAP analysis was performed to visualize the distribution of the 426,461 CpGs after dimension reduction. Each point in the UMAP represents a single CpG site, characterized by 60 β-values. CpG sites that show low methylation in both tumor and normal tissues are predominantly located at one extreme of the UMAP plot. Conversely, CpG sites that show high methylation in both tissues are found at the opposite extreme. Points situated in the central region of the UMAP correspond to CpG sites with a mixture of high and low β-values, reflecting differential methylation patterns across samples. Notably, the 160,587 significantly differentially methylated CpG sites identified through DMA are predominantly located in this central region (Fig. 3.5).

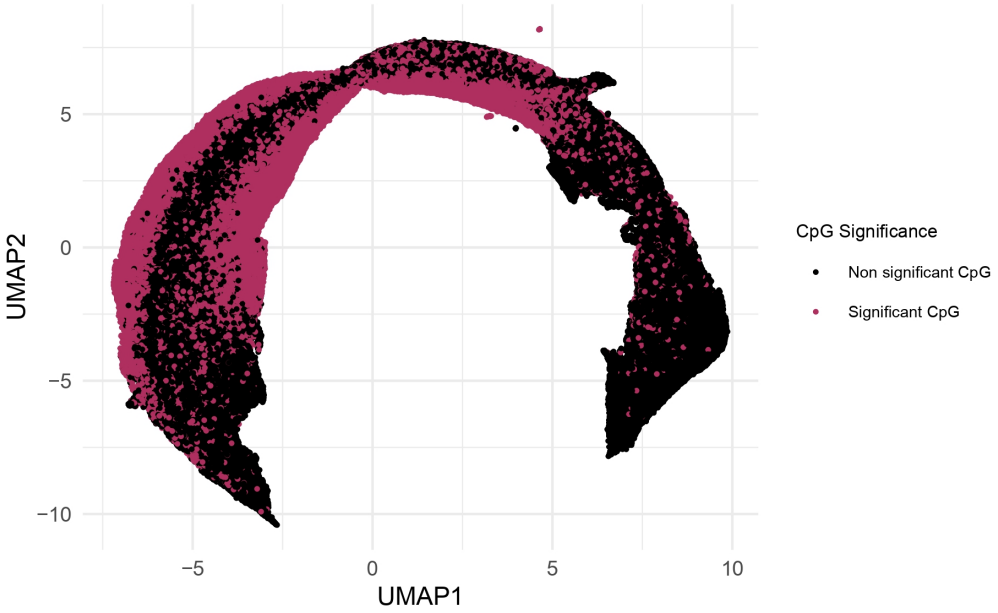


Figure 3.5: UMAP analysis of CpG distribution. Not significantly differentially methylated CpGs are colored in black, and significantly differentially methylated CpGs are colored in pink.

3.3 Differential methylation analysis of regions

While a probe-wise analysis is useful and informative, it is also interesting to explore whether several proximal CpGs are concordantly differentially methylated. For this differentially methylated regions should be identified.

For the DMR analysis, the top 5000 CpG sites were utilized and 642 significant regions were identified. Figure 3.8 presents a customized visualization of the most significant DMR. At the top, chromosome six is depicted with a red line indicating the position of the DMR. The red bar provides detailed information about the length of the region, with a scale for orientation on the chromosome. Below, additional information is provided about the methylation levels for the individual CpG sites, distinguishing between primary solid tumor and normal solid tissue. It can be observed that across all CpG sites, the primary solid tumor samples exhibit higher methylation levels than the normal solid tissue samples (Fig. 3.6).

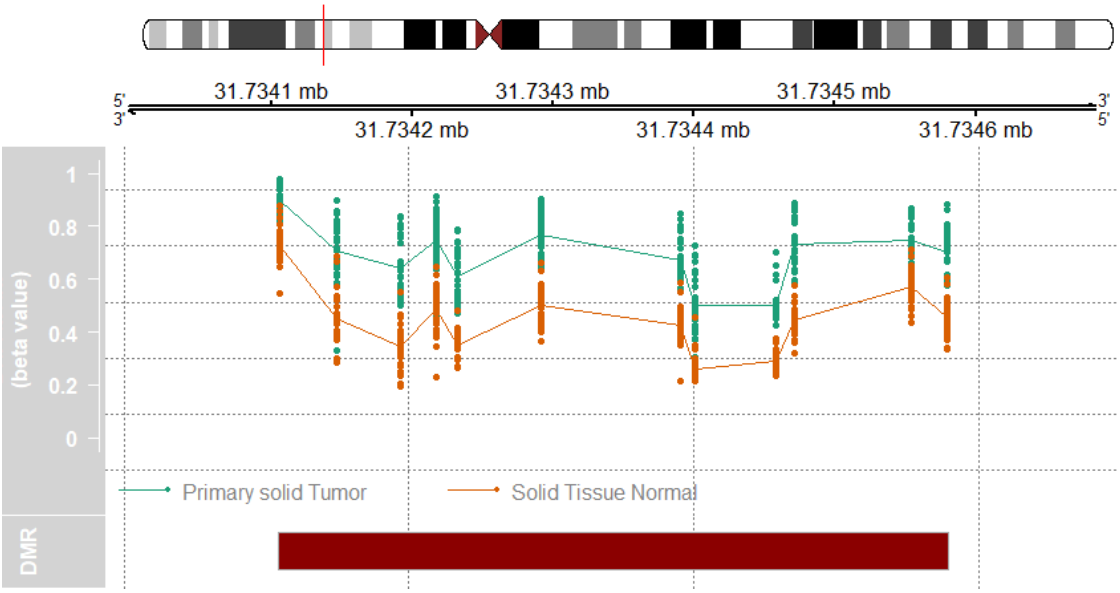


Figure 3.6: Customised visualization of the most significant DMR. The DMR is located on chromosome six. Below, the different CpG loci are shown with their methylation level.

3.4 Comparing results with current literature

3.4.1 Analysing diagnostic markers of (Zhang *et al.*, 2020)

The methylation patterns of seven diagnostic CpG sites (cg20383521, cg09804858, cg06741896, cg01668352, cg10708955, cg06998282, cg04658021), identified by Zhang *et al.*, were examined in both BRCA tumor and normal tissue samples.

Significant differences in methylation were observed between tumor and healthy tissue across all seven CpG sites. Specifically, cg20383521, cg09804858, and cg06741896 displayed hypomethylation in BRCA tumor tissue compared to healthy tissue (Adjusted p-values: 8.34×10^{-9}, 6.99×10^{-8}, and 7.98×10^{-4}, respectively). In contrast, cg01668352, cg10708955, cg06998282, and cg04658021 showed hypermethylation in BRCA tumor tissue (Adjusted p-values: 1.71×10^{-11}, 3.79×10^{-10}, 1.02×10^{-12}, and 1.72×10^{-9}, respectively)(Fig. 3.7A).

In the study, a high correlation between the seven markers was uncovered, implying that these differentially methylated sites could collectively influence the development of BRCA.

A correlation analysis between the CpGs in the methylation data also indicated strong correlations among the seven CpG sites (Fig. 3.7B).

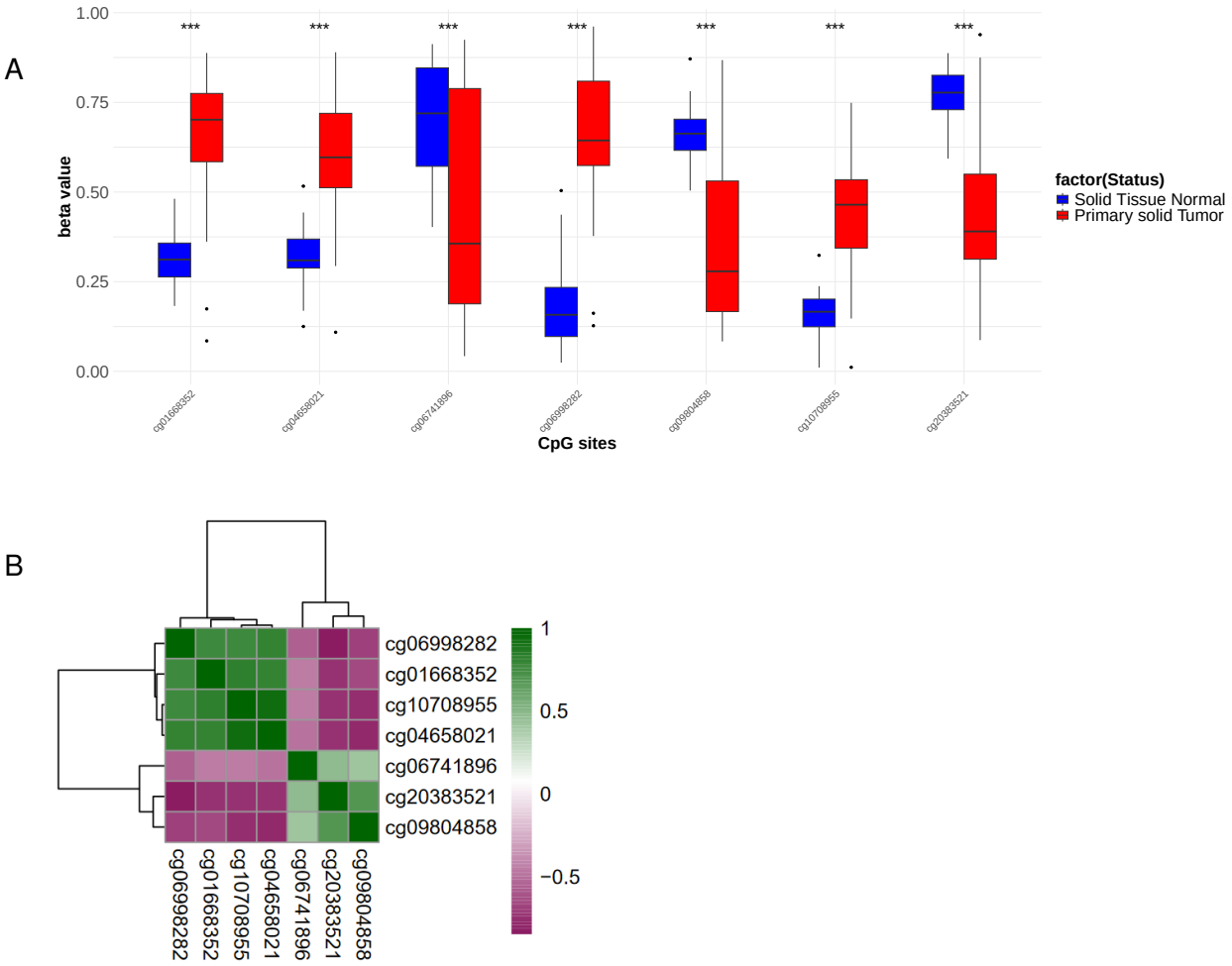


Figure 3.7: Methylation of seven diagnostic markers. (A) Boxplots of β-values of the CpGs. Normal samples are colored in blue and tumor samples are colored in red. Asterisks denote significance: *** $p < 0.001$. (B) Correlation heatmap of the diagnostic CpGs. Positive correlation is colored in green and negative correlation os colored in purple.

3.4.2 Comparing DMR results with top 250 genes of Panagopoulou *et al.*

In order to validate the DMR results of the given data, they were compared to the top 250 genes identified by Panagopoulou *et al.* in 2021. First, all 11,782 identified genes were compared to the top 250 genes of the study. An overlap of 149 genes was observed, indicating that 59.6% of their top 250 ranked genes were also present in the dataset analyzed (Fig. 3.8A).

To refine the analysis further, the intersection of the top 500 genes from the DMR results with the top 250 genes of the study was examined. This intersection revealed an overlap of 64 genes, showing that 25.6% of their top 250 genes were among the top 500 genes identified (Fig. 3.8B).

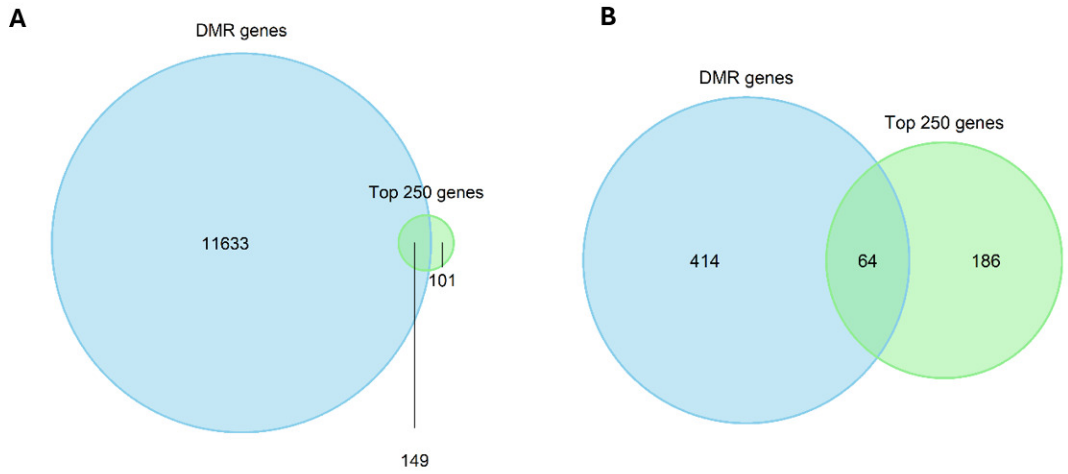


Figure 3.8: Comparing DMR results with top 250 genes of Panagopoulou *et al.* in 2021. (A) Venn diagram of 11,782 genes from DMR results compared with top 250 ranking genes of the paper. (B) Venn diagram of top 500 genes DMR compared with top 250 ranking genes of the paper.

3.5 Hallmark gene set enrichment analysis

The 160,587 significant CpG sites were used for hallmark gene set enrichment analysis from the Molecular Signature Database (MSigDB) (Liberzon *et al.*, 2015) .
The 10 most significant enriched hallmarks are the early response to estrogen, the epithelial-mesenchymal transition, myogenesis, apical junction, KRAS-mediated downregulation, inflammation response, IL-2 / STAT5, IL-6 / JAK / STAT3 signaling, late response to estrogen and allograft rejection (Fig. 3.9).

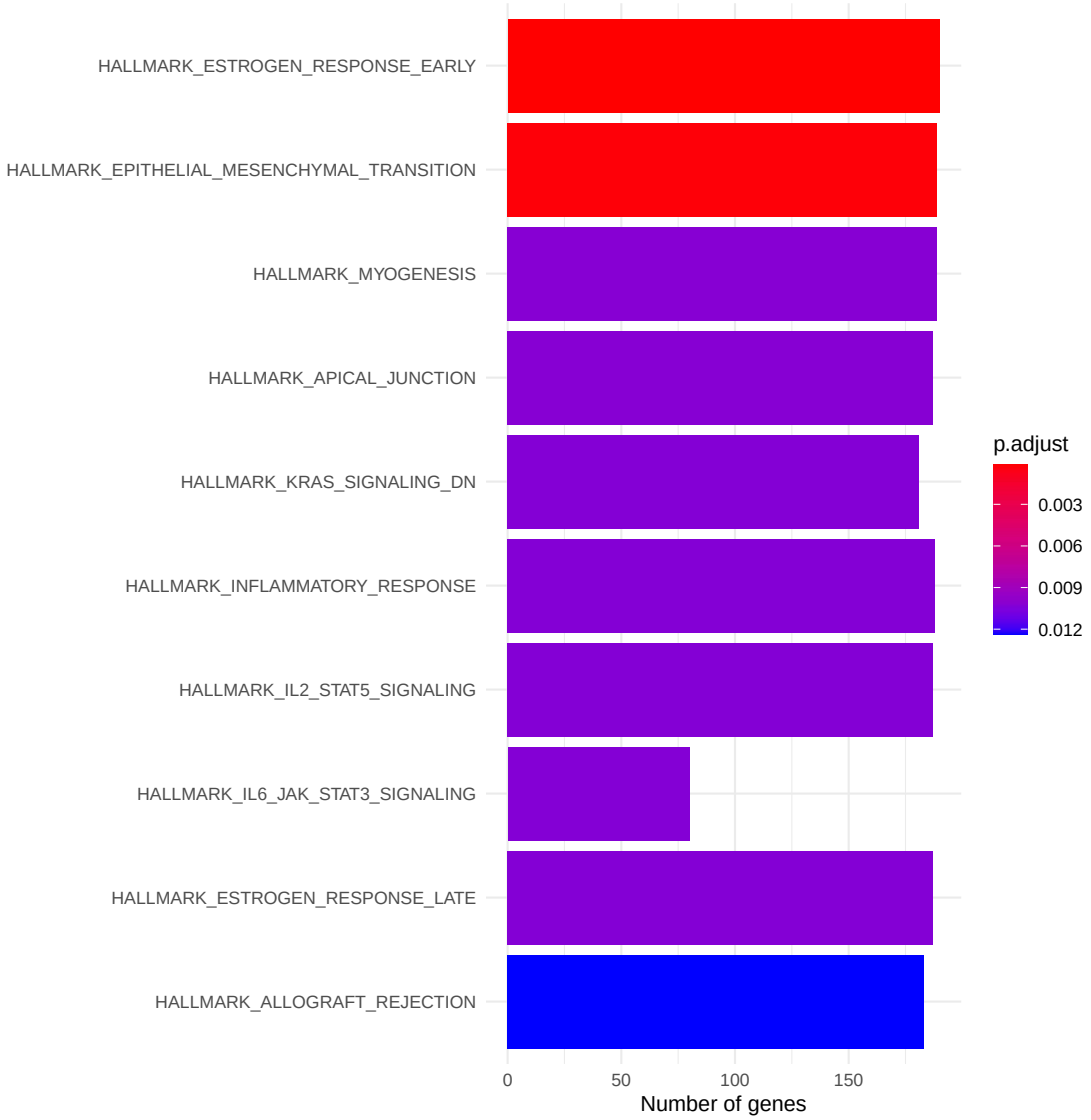


Figure 3.9: Bar plot showing the 10 most significantly enriched hallmarks. The hallmarks are sorted by their significance and are color-coded accordingly. Additionally, the x-axis displays the number of genes that constitute each hallmark.

4 Discussion

In this study, we performed an in-depth analysis of DNA methylation patterns in BRCA and normal tissues using data from The Cancer Genome Atlas. Utilizing the extensive data provided by the Illumina Infinium Human Methylation 450 BeadChip array, our objective was to identify specific CpG sites that showed differential methylation between tumor and normal samples. This approach allowed us to explore the epigenetic mechanisms that contribute to BRCA pathogenesis.

The results from the PCA clustering patients clearly show that normal solid tissue samples cluster more closely than the primary solid tumor samples with two tumor samples tightly within the normal cluster. This shows that not all tumors are necessarily epigenetically prominent and that genetic mutations are also sufficient for tumorigenesis. The cluster of tumor samples is much wider and poorly defined, showing a great epigenetic diversity among patients.

We performed a DMA to identify and characterize differentially methylated CpG sites associated with tumor tissues compared to normal tissues. Two methods were applied for this analysis and all 160,587 CpGs that were significant for both methods were used for downstream analysis. Visualization of the six most significantly methylated CpGs showed clear differences in methylation levels between tumor and normal tissues. Each CpG site pair exhibited either hypermethylation or hypomethylation in tumor tissues, highlighting the potential role of these epigenetic modifications in tumorgenesis.
Unfortunately, we do not have more information about the hypomethylated CpGs except that they are promoter associated. However, for the hypermethylated CpGs we have information about the name and genomic categories from the UCSC Reference Genome Browser Database (Karolchik *et al.*, 2004). The most significant CpG site is associated with methylation in the 5' UTR of the JAK1 gene. This gene encodes Janus kinase 1, which is pivotal in the IL-6/JAK1/STAT3 signaling pathway, crucial for immune and inflammation responses. This signaling pathway also emerged as a significant hallmark in our hallmark gene set enrichment analysis. Additionally, it plays a critical role in epithelial remodeling and metastasis, both of which are major factors in cancer progression (National Institutes of Health, 2024a). The fifth

and sixth most significant CpGs are both associated with hypermethlation in the gene body. The fifth CpG is associated with PGCP, which provides a major component of the gastric mucosa (National Institutes of Health, 2024b). According to Gao *et al.*, hypermethylation of PGCP is linked to the immortalization of bronchial epithelial cells and lung cancer (Gao *et al.*, 2018). Immortalization is a critical step in tumor development as it allows for unconstrained cell proliferation (Counter *et al.*, 1998) and is listed in the hallmarks of cancer by Hanahan and Weinberg. Consequently, it could also play a role in BRCA formation. The sixth CpG site is associated with ZYG11B, a cell cycle regulator involved in protein quality control, specifically in proteasomal ubiquitin-dependent protein degradation processes (National Institutes of Health, 2024c). It is believed to contribute to chemoresistance in colorectal cancer by inducing autophagy (Duan *et al.*, 2023). Dysregulated autophagy is associated with tumorigenesis and resistance to cancer therapies (X. Li *et al.*, 2019) and could therefore also be important in BRCA pathogenesis.

When examining the significantly differentially methylated CpGs, it becomes clear from both the heatmap of the correlation among the top 1000 significantly differentially methylated CpGs and the PCA of the top 1000 significantly differentially methylated CpGs that there is a clear separation and clustering between hypermethylated and hypomethylated CpG sites. The strong correlation and clustering of the CpGs based on their methylation status suggest that these methylation changes are functionally relevant. These epigenetic modifications could play a role in the regulation of genes involved in tumor development and progression.
The UMAP plot reveals distribution of CpG sites based on their methylation patterns in tumor and normal tissues. CpG sites with consistently low or high methylation levels in both normal and tumor tissue are segregated to opposite extremes, while those displaying mixed β-values are centrally located. Notably, the significant differentially methylated CpG sites identified through DMA occupy the central region, reflecting their significant methylation differences between tumor and normal tissues.
When comparing patient clustering based on tumor type before and after the DMA, focusing solely on significantly different methylated CpGs notable differences in clustering patterns are highlighted. Post-DMA, clusters exhibit greater distinctiveness, with patients distinctly aligning according to their tumor type within one of these clusters. This indicates that these significant CpGs could serve as potential diagnostic markers. Future research could focus on validating these CpGs as biomarkers and exploring their functional roles in cancer development and progression.

While probe-wise analysis offers valuable insights, it is also important to investigate whether clusters of proximal CpGs exhibit concordant differential methylation. To

address this, we performed a DMR analysis using the top 5000 CpG sites. This analysis produced a detailed table of DMRs sorted by significance and annotated with chromosome location, range, associated genes, and other relevant parameters. Identifying DMRs provides insight into broader epigenetic changes that may influence cancer-related gene regulation.

The customized visualization of DMRs highlighted a significant region on chromosome six. It can be clearly seen that primary solid tumor samples consistently exhibited higher methylation levels across all CpG sites in this region compared to normal tissues. These findings indicate that hypermethylation is a common feature in tumor tissues, which may contribute to gene silencing and tumor progression. The most significant DMR is associated with the VWA7 gene, which encodes the von Willebrand factor A domain-containing protein 7. While the von Willebrand factor is typically linked to blood coagulation, the von Willebrand factor A (VWA) domain is also present in other proteins, including extracellular proteins involved in cell adhesion, extracellular matrix proteins, integrin receptors, and intracellular proteins involved in transcription, DNA repair, transport, and the proteasome. At a higher level, it appears to be particularly involved in protein-protein interactions within multiprotein complexes (Whittaker and Hynes, 2002). All these functions can be relevant for cancer biology. It is also thought to contribute to lung cancer susceptibility (Kumánovics and Lindahl, 2001). However, there is no established link to BRCA. This potential connection warrants further investigation in future research studies.

The significance and high correlation of the seven diagnostic markers identified by Zhang et al. (2020) in the given data show that our results align with existing literature, reinforcing the validity and relevance of the presented results. Furthermore, the correlation could suggests that these CpG sites may collectively influence BRCA pathogenesis. To confirm this hypothesis, these seven diagnostic CpGs need to be studied in greater detail, including their localization and influence on pathways.
The overlap observed between our DMRs and Panagopoulou *et al.* 250 top-ranked genes, highlights the robustness and reproducibility of our findings across independent datasets. The fact that the genes do not overlap completely is partly due to the different DMA methods used. Additionally, differences in sample size and data processing and normalization techniques can also influence the CpG methylation results. This validation not only reaffirms the reliability of our DMR analysis but also reinforces the importance of these genomic regions in BRCA pathogenesis which could be further investigated.

Furthermore, the 160,587 significant CpG sites were enriched with the hallmark gene set from the Molecular Signature Database. These hallmarks represent critical biological

pathways and processes that could provide a deeper understanding of the molecular mechanisms driving BRCA. In particular, the IL-2/STAT5 signaling pathway and the Estrogen Response pathways are of significant interest in the pathogenesis of BRCA. STAT5 is a protein that is commonly associated with cancer, especially with BRCA. In early stages of BRCA pathogenesis, it appears to promote tumor growth, yet in later stages, it helps with differentiation of mammary epithelium. Activation of STAT5 is associated with favourable clinical outcomes (Tan and Nevalainen, 2008). Further analysis could entail grouping our samples according to cancer stage and analysing whether methylation status of genes affected by STAT5 differs by cancer stage. Their methylation level could be of therapeutic significance since treatment for STAT5-mutated tumours is available (Cumaraswamy *et al.*, 2012).
More than two-thirds of breast cancer patients express estrogen receptor-α (ER), making it an important target for therapeutic strategies. ER activation initiates diverse downstream phenotypic cascades within BRCA cells. This includes the regulation of estrogen response genes, which emerge as significant hallmark gene sets from the gene enrichment analysis.
Several studies associate estrogen response genes with breast cancer development. Early estrogene response genes are enriched in functions related to canonical ER signaling and transcriptional activation mediated by RNA polymerase II. Late estrogen response genes are enriched in developmental pathways, metabolic processes and GPCR signaling, which are essential for survival and growth of BRCA cells (Z. Li *et al.*, 2023). Notably, elevated expression of estrogen response genes, both early and late is linked to heightened risk of late recurrence in ER+/HER2- breast cancer patients (Bense *et al.*, 2017).
These findings show that the hallmarks which result from the methylation patterns of the given data are strongly associated with breast cancer, further supporting the relevance in using DNA methlation patterns as possible diagnostic, prognostic and therapeutic approaches. Unfortunately, it is unknown whether the patients also had metastases. Many of the hallmarks identified in this analysis are associated with metastasis, such as EMT dysregulation. Further research could analyse this and develop DMA and DMR analysis into a prognostic tool for metastasis.
In conclusion, this hallmark gene set enrichment analysis shows that the genes identified in the project are associated with cancer and breast cancer.

Our comprehensive analysis of DNA methylation patterns in BRCA pathogenesis reveals distinct epigenetic modifications that differentiate tumor tissues from normal tissues. Furthermore, the Differential Methylation Analysis (DMA) employed in this study not only identified specific CpG sites but also provided insights into broader epigenetic changes that may influence cancer-related gene regulation. The identification of hypermethylated CpGs associated with genes such as JAK1, PGCP, and ZYG11B underscores their potential roles

in critical cancer-related pathways, including immune response, cell immortalization, and chemoresistance. The clear clustering of methylation profiles post-DMA suggests these modifications are functionally significant and may serve as valuable diagnostic markers. The enrichment of hallmark gene sets related to cancer, including those associated with estrogen response, further emphasizes the relevance of these methylation patterns. Importantly, our findings align with existing literature, reinforcing the robustness of our results. This project highlights DMA as a powerful tool for unraveling the complex epigenetic landscape of breast cancer and underscores its potential for guiding personalized therapeutic strategies.

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